

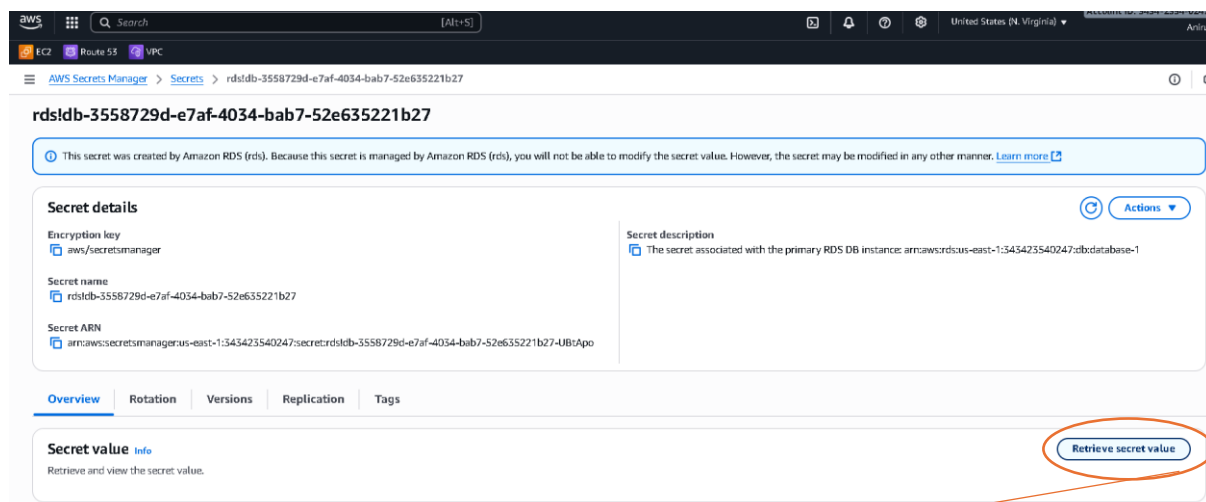
RDS Lab :

Task 1 – To create a Database Instance By using AWS Secret Manager .

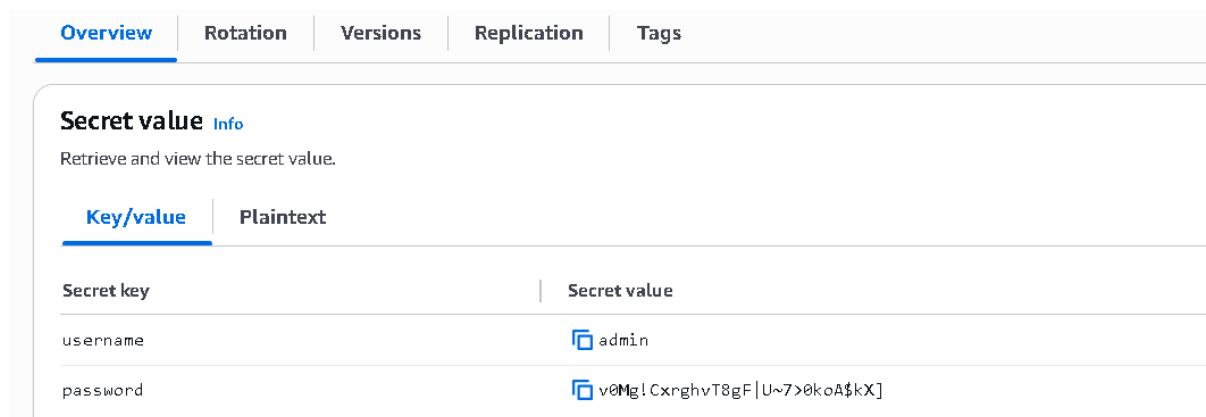
First create a Database Instance and there is one option to select AWS secret Manager and Self Managed .

So click to AWS secret Manager and the automated Rotation secret key is created for the DB instance.

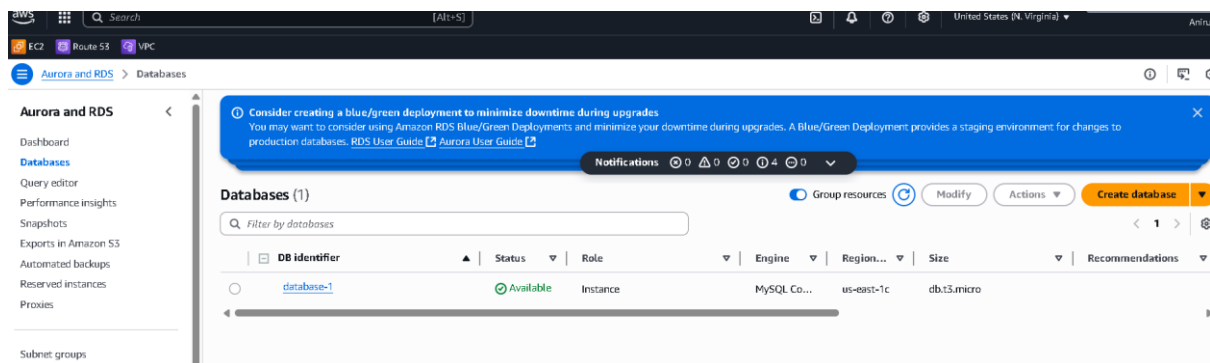
Here rotation means that after a certain period of time which we give, the key changes automatically and new key generates.



Here is the new secret, By clicking this we can see the username and password by which we can connect with the MYSQL workbench.



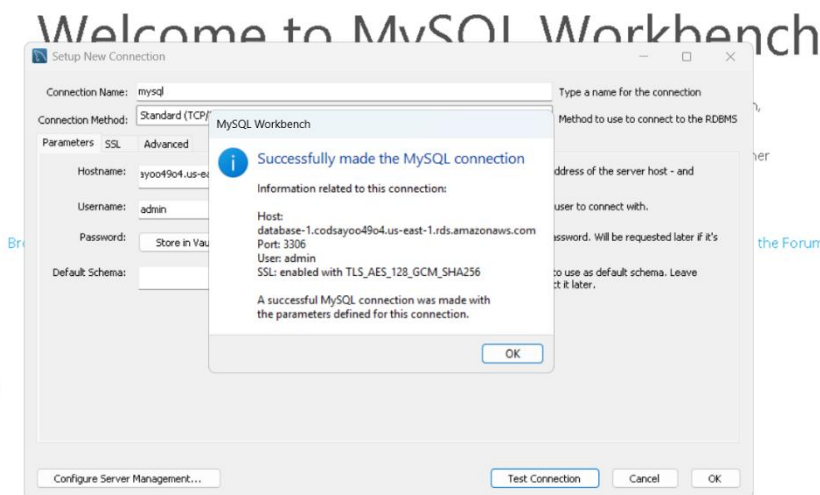
Now the Database Instance is created.



By this we can connect with the mysql workbench . Before that we need to check :-

1. we have created the DB instance in public.
2. We have to give security group a inbound rule of MYSQL/Aurora.

After this only we can connect to the Mysql workbench.



Give the configurations , hostname, username, password (from AWS secret manager).

Now once connected , run the query

1. Create database
2. Create table
3. Insert the columns and value
4. Select the whole table to show the table.

I have written down the query , you can copy and paste to the sql editor.

```
CREATE DATABASE IF NOT EXISTS SimpleSchoolDB;
USE SimpleSchoolDB;
CREATE TABLE Students (
  id INT PRIMARY KEY AUTO_INCREMENT,
  student_name VARCHAR(100),
  subject VARCHAR(50),
  marks INT
);
INSERT INTO Students (student_name, subject, marks) VALUES
('John Doe', 'Mathematics', 85),
('John Doe', 'Science', 92),
('Jane Smith', 'Mathematics', 95),
('Jane Smith', 'History', 88),
('Peter Jones', 'Science', 76);
SELECT * FROM Students;
```

You can also give time window for maintenance , backups , so that this operations performed it that time only , so it doesn't effect the application or database.

Modify DB instance: database-1

Summary of modifications

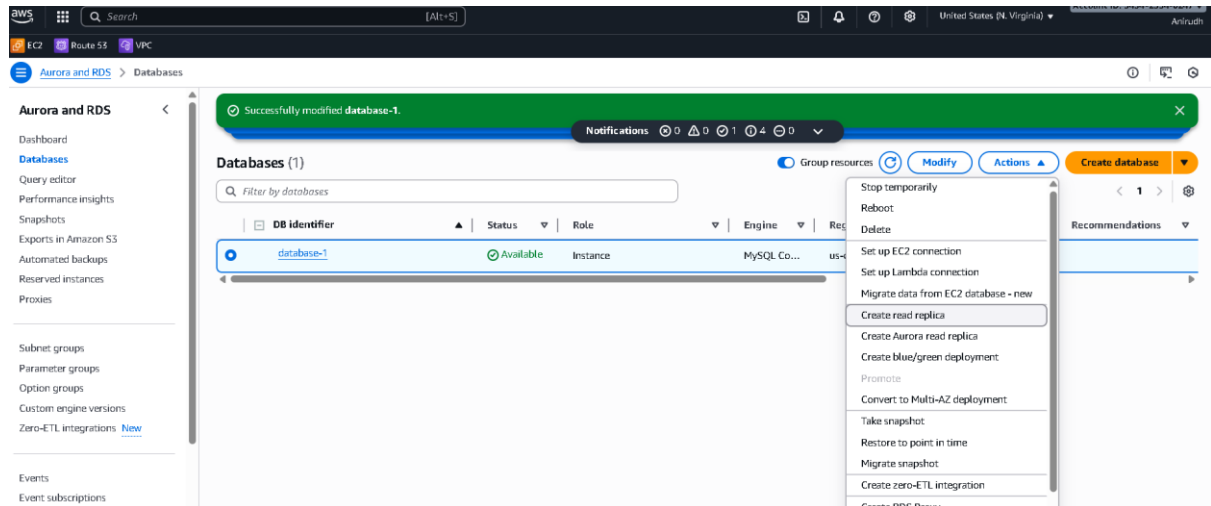
You are about to submit the following modifications. Only values that will change are displayed. Carefully verify your changes and choose **Modify DB Instance**.

Attribute	Current value	New value
Manage master credentials	Yes	No
Master password		*****
Backup window	06:14-06:44	06:00-06:30
Maintenance window	mon:03:57-mon:04:27	mon:03:00-mon:03:30

By **Modify** option in database instance we can modify the instance to change the configuration.

Task 2. Creating a Replica of the Database and check that the database is there or not.

So we already have the database instance , so click on action and there is a option to create a read replica.



By this the read replica is created but we can only read it , not change anything from the database.

Same connect the instance with the replica configuration.

But here while using we have to change the instance to Self managed.

Aws secret manager wont work on replicas.

So create a password and then connect the instance to Mysql workbench.

Welcome to MySQL Workbench

MySQL Workbench is the official graphical user interface (GUI) tool for MySQL. It allows you to design, create and browse your database schemas, work with database objects and insert data as well as design and run SQL queries to work with stored data. You can also migrate schemas and data from other database vendors to your MySQL database.

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MySQL Connections

Local instance MySQL80 root localhost:3306	mysql admin database-1.codsayoo49o4.us-east-1...	replica admin replica-db.codsayoo49o4.us-east-1.r...
--	--	--

Now check the data is there or not.

The screenshot shows the MySQL Workbench interface. On the left, the 'Navigator' pane displays the 'SimpleSchoolDB' schema with a table named 'Students'. The 'Columns' pane shows the table structure: id (int), student_name (varchar(100)), subject (varchar(50)), and marks (int). The 'Query Editor' shows a query: `USE SimpleSchoolDB;` followed by `SELECT * FROM Students;`. The 'Result Grid' displays the following data:

id	student_name	subject	marks
1	John Doe	Mathematics	85
2	John Doe	Science	92
3	Jane Smith	Mathematics	95
4	Jane Smith	History	88
5	Peter Jones	Science	76

The 'Output' pane shows the execution log with three steps: 1. 12:53:03 SELECT * FROM Students LIMIT 0, 1000 (0 row(s) affected), 2. 12:53:53 USE SimpleSchoolDB (5 row(s) returned), and 3. 12:54:08 SELECT * FROM Students LIMIT 0, 1000 (5 row(s) returned).

Now delete the Main database instance , by doing this the replica is become the main instance, now from this instance we can create the database and edit also. Now this instance have read and write both the permission.

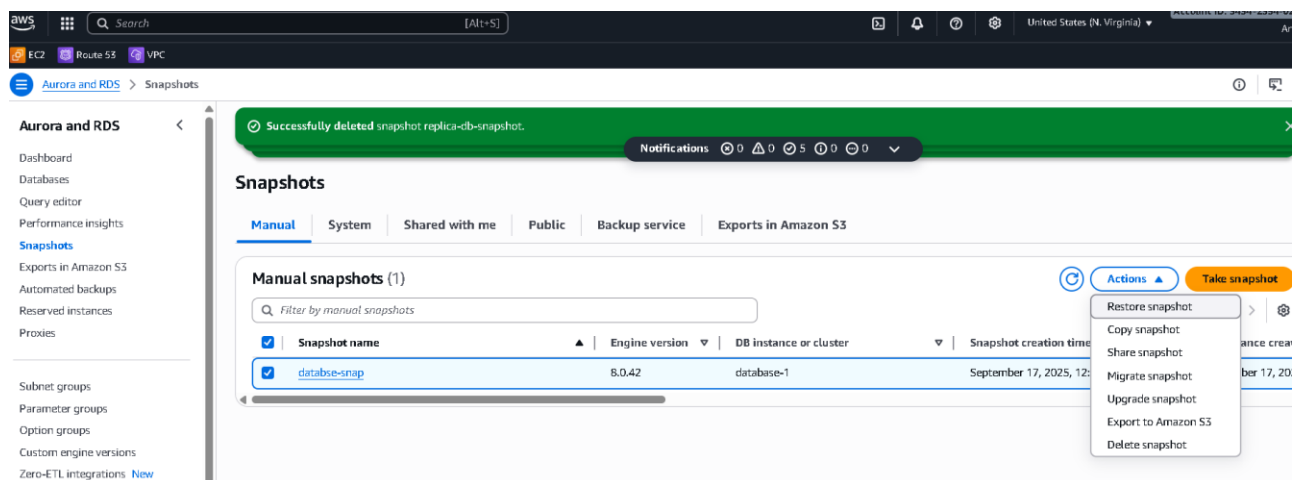
Now the main difference between the main instance this replicated instance , is that the replicated instance cannot create another read replica.

The screenshot shows the AWS Management Console 'Aurora and RDS' section. A notification at the top states 'Successfully deleted DB instance database-1'. Below, the 'Databases (1)' table shows a single instance named 'replica-db' with status 'Available'. A context menu is open for this instance, showing various actions. A text box with an arrow points to the 'Create read replica' option in the menu.

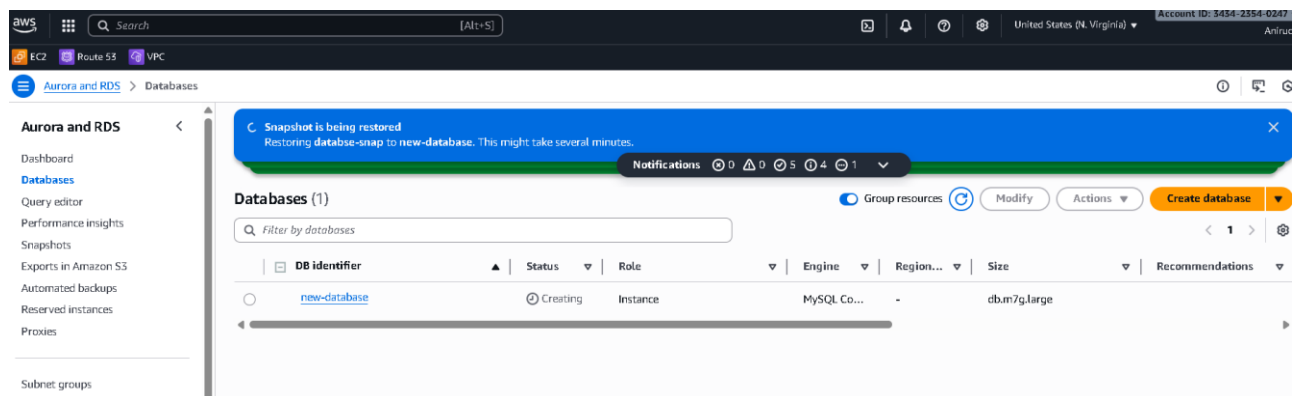
See you don't have the option of "Create read replica"

Task 3. Create snapshot of the DB instance and from that snapshot, create a new DB instance.

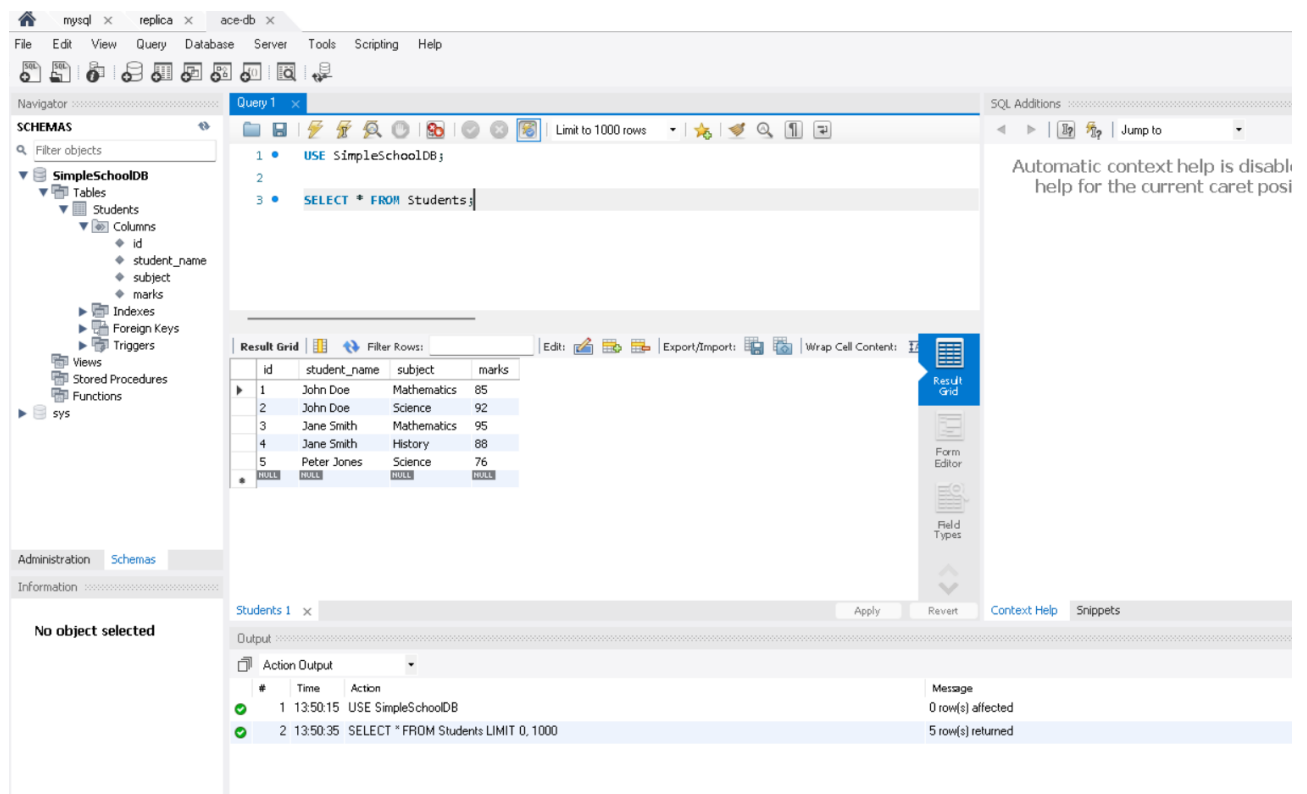
As I already created the database instance , now create a snapshot.



Now from this snapshot by clicking the “Restore snapshot” we can create the new database instance.



Now you can check that the database and tables is there in the new database instance .

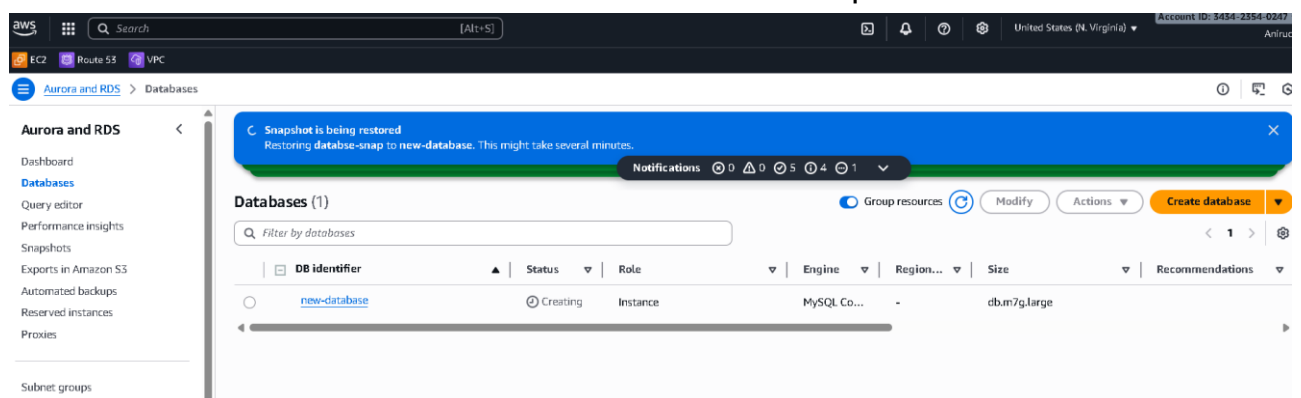


By this you can see the data is on the instance.

Task 4. To access the Private DB Instance with the help of EC2 instance.

First we have to create a DB instance with configuration of 'Not giving public access'.

Well, I had modify the snapshot restored Instance with clicking no public access. You can create another DB instance without public access.



After this we have to create a EC2 instance in public and add key while creating.

Now connect the instance, now run the following commands :-

`sudo su -`

`yum install mariadb105-server -y`

`mysql -h <hostname> -u <admin> -p`

By this you can run query of creating database, table ,etc in instance.

This is how you can access the Mysql database from EC2 instance.



```
#####
      _               _
     _/_/            _/_/
    _/_/            _/_/
   _/_/            _/_/
  _/_/            _/_/
 _/_/            _/_/
/_/            _/_/

#####
Amazon Linux 2023

https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-31-16-178 ~]$ sudo su -
[root@ip-172-31-16-178 ~]# yum install mariadb-server -y
Amazon Linux 2023 Kernel Livepatch repository
No match for argument: mariadb-server
Error: Unable to find a match: mariadb-server
[root@ip-172-31-16-178 ~]# yum install mariadb105-server -y
Last metadata expiration check: 0:01:10 ago on Wed Sep 17 08:08:28 2025.
Dependencies resolved.
=====
Package                               Architecture      Version            Repository          Size
=====
Installing:
mariadb105-server                     x86_64            3:10.5.29-1.amzn2023.0.1  amazonlinux         10 M
Installing dependencies:
mariadb-connector-c                   x86_64            3.3.10-1.amzn2023.0.1    amazonlinux          211 k
mariadb-connector-c-config            noarch            3.10.5.29-1.amzn2023.0.1  amazonlinux           9.9 k
mariadb105                            x86_64            3:10.5.29-1.amzn2023.0.1  amazonlinux          1.5 M
mariadb105-common                     x86_64            3:10.5.29-1.amzn2023.0.1  amazonlinux           28 k
mariadb105-errmsg                     x86_64            3:10.5.29-1.amzn2023.0.1  amazonlinux          212 k
mysql-selinux                         noarch            1.0.4-2.amzn2023.0.3     amazonlinux           36 k
perl-pod                              x86_64            1.80-477.amzn2023.0.7    amazonlinux           177 k
=====
```



```
perl-DBD-MariaDB-1.22-1.amzn2023.0.4.x86_64    perl-DBI-1.643-7.amzn2023.0.3.x86_64          perl-Data-Dumper-2.174-460.amzn2023.0.2.x86_64
perl-File-Copy-2.34-477.amzn2023.0.7.noarch     perl-FileHandle-2.03-477.amzn2023.0.7.noarch    perl-Math-BigInt-1.1.9998.39-2.amzn2023.0.2.noarch
perl-Math-BigRat-0.2624-500.amzn2023.0.2.noarch perl-Math-Complex-1.59-477.amzn2023.0.7.noarch  perl-Sys-Hostname-1.23-477.amzn2023.0.7.x86_64
perl-base-2.27-477.amzn2023.0.7.noarch

Complete!
[root@ip-172-31-16-178 ~]# mysql -h new-database.codsayoo49o4.us-east-1.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 12
Server version: 8.0.42 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> CREATE DATABASE IF NOT EXISTS SimpleSchoolDB;
Query OK, 1 row affected (0.009 sec)

MySQL [(none)]> USE SimpleSchoolDB;
Database changed
MySQL [SimpleSchoolDB]> CREATE TABLE Students (
  ->   id INT PRIMARY KEY AUTO INCREMENT, -- A unique ID for each row
  ->   student_name VARCHAR(100),         -- The student's full name
  ->   subject VARCHAR(50),               -- The subject for the mark
  ->   marks INT                          -- The marks they received
  -> );
Query OK, 0 rows affected (0.027 sec)

MySQL [SimpleSchoolDB]> INSERT INTO Students (student_name, subject, marks) VALUES
  -> ('John Doe', 'Mathematics', 85);
```



```
aws
[Search] [Alt+S]
EC2 Route 53 VPC

mysql [SimpleSchoolDB]> CREATE TABLE Students (
-> id INT PRIMARY KEY AUTO_INCREMENT, -- A unique ID for each row
-> student_name VARCHAR(100), -- The student's full name
-> subject VARCHAR(50), -- The subject for the mark
-> marks INT -- The marks they received
-> );
Query OK, 0 rows affected (0.027 sec)

mysql [SimpleSchoolDB]> INSERT INTO Students (student_name, subject, marks) VALUES
-> ('John Doe', 'Mathematics', 85),
-> ('John Doe', 'Science', 92),
-> ('Jane Smith', 'Mathematics', 95),
-> ('Jane Smith', 'History', 88),
-> ('Peter Jones', 'Science', 76);
Query OK, 5 rows affected (0.006 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql [SimpleSchoolDB]> SELECT * FROM Students;
+-----+-----+-----+-----+
id | student_name | subject | marks |
+-----+-----+-----+-----+
1 | John Doe | Mathematics | 85 |
2 | John Doe | Science | 92 |
3 | Jane Smith | Mathematics | 95 |
4 | Jane Smith | History | 88 |
5 | Peter Jones | Science | 76 |
+-----+-----+-----+-----+
5 rows in set (0.001 sec)

mysql [SimpleSchoolDB]>
```

i-085918ae584e1f060 (ace-db-server)
PublicIPs: 34.229.125.184 PrivateIPs: 172.31.16.178

CloudShell Feedback

Now in MYSQL workbench to connect this DB instance you have to change connection method from TCP/IP to standard TCP/IP over SSH. Now fill the configuration like , ssh hostname(public ip), ssh name(ec2-user), private key file location, and the DB instance configuration.

Manage Server Connections

MySQL Connections

- Local instance MySQL80
- mysql
- replica
- ace-db**

Connection Name: ace-db

Connection Remote Management System Profile

Connection Method: Standard TCP/IP over SSH Method to use to connect to the RDBMS

Parameters SSL Advanced

SSH Hostname: ec2-34-229-125-184.compute-1.amazonaws.com SSH server hostname, with optional port number.

SSH Username: ec2-user Name of the SSH user to connect with.

SSH Password: Store in Vault ... Clear SSH user password to connect to the SSH tunnel.

SSH Key File: C:\Users\Anirudh\Downloads\ace-key. ... Path to SSH private key file.

MySQL Hostname: new-database.codsayoo49o4.us-east-1.rds.amazonaws.com MySQL server host relative to the SSH server.

MySQL Server Port: 3306 TCP/IP port of the MySQL server.

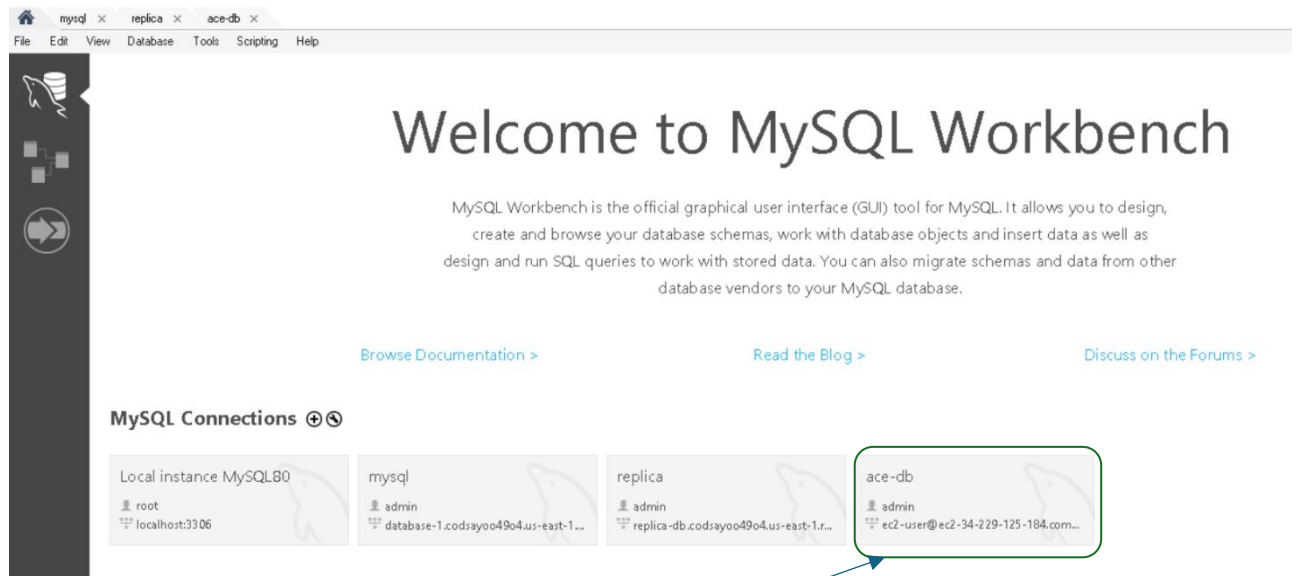
Username: admin Name of the user to connect with.

Password: Store in Vault ... Clear The MySQL user's password. Will be requested later if not set.

Default Schema: The schema to use as default schema. Leave blank to select it later.

New Delete Duplicate Move Up Move Down Test Connection Close

Now you see the Private DB instance is connected to MYSQL workbench via EC2 instance.



Now you can check the data in the SQL editor.